

DevNexus | AgileBridge

SRS

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 agilebridge
CONSIDERED
SOFTWARE SOLUTIONS

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1.Introduction

Overview

UniTrade is a peer-to-peer marketplace platform built for South African university students to buy and sell items such as textbooks, study materials and other students essentials. The platform verifies student identity before granting full access, uses AI-assisted risk scoring to screen listings before they go live and manages the entire transaction lifecycle, from browsing and reservation, through in-person meetups, to secure payment and handover confirmation. Payments are processed via Payfast and PIN-based system confirms that a handover has taken place. A reputation and dispute-handling system supports trust between users.

Key System Features

- **Student verification** – Sign-up with university email domain checking, OTP verification and proof-of-registration upload, backed by an AI scanning layer with admin fallback for uncertain cases.
- **AI-assisted listing moderation** – Listings are automatically risk-scored (reserve-image checks, content classification) and routed to go live or held for admin review.
- **Marketplace browse** – View listings with title, price, condition, category and photos, reserve and add to your wish list directly from the product card.
- **Reservations** – Express interest in an item to trigger an automated message and a time-boxed reservation, with clear acknowledgement and response windows.
- **In-app chat** – Coordinate directly with the other party to arrange meetup time and location through a meetup scheduler.
- **Secure payment** – Pay instantly via Payfast after inspecting the item in person.
- **PIN-based handover confirmation** – A PIN generated after payment confirms the item changed hands.
- **Ratings and reputation** – Rate the other party after a transaction, reputation scores factor into trust and abuse detection.
- **Dispute reporting** – flag listing-quality issues, no-shows, or false listing claims for admin review.
- **Recommendations** – Behaviour-driven suggestions like “Popular for course code” or “Frequently bought together”, based on real student activity rather than a fixed material list.

2. Epics, User Stories and Acceptance Criteria

Epic 1: User and Identity Management

Enable South African university students to register, authenticate, and verify their student status so that the marketplace remains exclusive to legitimate students.

US-1.1 — Register an Account

As a: prospective student user I want to register with my name, surname, university, degree, year of study, student email, and password so that I can begin the process of accessing the marketplace.

Acceptance Criteria:

- Given I am on the registration screen, when I submit valid details with a recognised SA university email domain, then my account is created in pending verification state.
- Given I submit an email with a non-SA-university domain, when I attempt to register, then registration is rejected with a clear message naming the domain check as the reason.
- Given I submit an email already registered, when I attempt to register, then I see a message prompting me to log in or reset my password (the system does not reveal whether the account is verified).

Given I am on the registration screen, when I submit the form with missing or invalid required fields, then I see field-level error messages and the form is not submitted.

US-1.2 — Receive and Verify OTP

As a: newly registering user I want to receive a one-time PIN on my student email and enter it so that the system confirms I control the email address.

Acceptance Criteria:

- Given my account was just created, when registration completes, then an OTP is sent to my student email within 60 seconds.
- Given I enter the correct OTP within its validity window, when I submit, then my email is marked verified and I proceed to the proof-of-registration step.
- Given I enter an incorrect OTP, when I submit, then I see an error and remain on the OTP screen; after a configured number of failed attempts, the OTP is invalidated and I must request a new one.
- Given the OTP has expired, when I submit it, then the system rejects it and offers to resend.
- Given I request a resend, when triggered, then a new OTP is sent and the previous one is invalidated; resends are rate-limited.

US-1.3 — Upload Proof of Registration

As a: verified-email user I want to upload my proof of registration as a PDF so that the system can verify, I am a current student.

Acceptance Criteria:

- Given I have verified my email, when I reach the upload step, then I can select a PDF file from my device.
- Given I attempt to upload a non-PDF file, when I submit, then the upload is rejected with a message stating PDF is required.
- Given the file exceeds the maximum size, when I submit, then the upload is rejected with a clear size limit message.
- Given a successful upload, when complete, then my proof is queued for AI verification and my status updates to 'under review'.

US-1.4 — AI-Assisted Proof Verification (System-Driven)

As the: system I want to automatically score uploaded proofs of registration for legitimacy so that confident-positive cases are auto-approved without admin workload.

Acceptance Criteria:

- Given a PoR has been uploaded, when AI scoring runs, then it returns a confidence score and a routing decision: auto approve, manual review, or manual review high risk.
- Given the score is in the auto-approve band, when scoring completes, then the user's account is upgraded to verified and the user is notified in-app and by email.
- Given the score is in the manual-review band, when scoring completes, then the case is added to the admin verification queue and the user remains in 'under review'.
- Given the score is in the manual-review-high-risk band (clear forgery indicators), when scoring completes, then the case is also routed to the admin queue with a high risk flag — the AI never auto-rejects; rejection is always a human decision.
- Given a user has uploaded a PoR, when AI scoring is in progress, then the user sees a loading indicator until a result is returned.
- Given a PoR has been scored, when the result is stored, then the system records the score, decision, timestamp, and model version for audit purposes.

US-1.5 — Manual Verification by Admin

As an: admin I want to review proofs of registration that AI flagged for manual review so that legitimate students get verified and forgeries are rejected.

Acceptance Criteria:

- Given a case is in the admin verification queue, when I open it, then I see the uploaded PoR, the user's submitted details, the AI score, and any AI-flagged reasons.
- Given I approve the case, when I confirm, then the user's account is upgraded to verified and they receive an in-app and an email notification.
- Given I reject the case, when I confirm, then I must select a terminal rejection reason (forged, expired, wrong document type, other) and add a free-text note; the user is notified and cannot resubmit without admin intervention.
- Given I request changes, when I confirm, then I must select a request changes reason (unreadable, missing information, other) and add a free text note; the user is notified with the reason and prompted to resubmit.
- Given I perform any verification action (approve, reject, request changes), when the action is confirmed, then the system records the admin ID, action taken, timestamp, and reason for audit purposes.
- Given a case has exceeded the defined SLA threshold, when I view the verification queue, then the case is prioritised and surfaced at the top of the queue.

US-1.6 — Resubmit Proof of Registration

As a: user whose PoR was rejected or required changes I want to upload a new PoR so that I can complete verification.

Acceptance Criteria:

- Given my PoR is in the 'changes requested' state, when I log in, then I see a clear prompt to resubmit and the reason for the previous rejection.
- Given my PoR is in the 'changes requested' state, when I check my emails, then I see a clear reason why my submission was declined and an instruction to resubmit.
- Given my PoR has been rejected after a manual admin review due to legitimacy concerns (e.g., forgery), when I attempt to access the application, then I am denied access and I receive both an in-app rejection notice and an email explaining the reason for the decision.
- Given my PoR is in the 'changes requested' state and the request was made by Admin A, when I resubmit a new PoR, then the resubmission is assigned to Admin A for review (not a different admin).

US-1.7 — View Verification Status

As a: registered user I want to see my current verification state at any time so that I know what I can and can't do on the platform.

Acceptance Criteria:

- Given I am logged in, when I open my account/profile area, then I can see my current state: pending email verification, pending PoR upload, under review, verified, rejected, or changes requested.
- Given I am in 'under review', when I view my status, then I see an estimated time remaining (based on SLA) and a note that drafts can be created but listings won't go live until verified, and I can browse listings.
- Given I am in 'rejected' or 'changes requested', when I view my status, then I see the reason and any next-step actions.

US-1.8 — Login

As a: registered student I want to log in with my student email and password so that I can access my account.

Acceptance Criteria:

- Given I enter correct credentials, when I submit, then I am authenticated and routed to the appropriate landing page based on my verification status.
- Given I enter incorrect credentials, when I submit, then I see a generic 'invalid email or password' message.
- Given I am logged in and idle for longer than the configured inactivity timeout (e.g. 30 minutes), when I attempt to access a protected page, then my session is expired and I am redirected to the login page.
- Given I check the 'remember me' box during login, when I close and reopen my browser or return after a long period (but within the extended session lifetime), then I am still required to re-authenticate.

US-1.9 — Reset Password

As a: student who has forgotten my password I want to reset it via an email-verified flow so that I can regain access.

Acceptance Criteria:

- Given I select 'forgot password', when I enter my registered email, then a password reset OTP is sent to my email.
- Given I submit a valid OTP within its validity window, when I enter a new password, then my password is updated and I am logged in.
- Given the OTP has expired or been used, when I attempt to use it, then it is rejected and I am prompted to start the reset flow again.

US-1.10 — Logout

As a: logged-in student I want to log out so that my session ends and my account remains secure on shared devices.

Acceptance Criteria:

- Given I am logged in, when I select logout, then I am successfully logged out of the system.
- Given I have logged out, when I attempt to access any authenticated route, then I am redirected to the login screen.
- Given I have logged out, when I try to perform any authenticated action, then it is rejected as unauthorised.

Epic 2: Listing and Inventory Management

Verified students can create, manage, and remove listings. Every submitted listing is auto scored by AI; high-risk listings route to admin review before going live. Unverified students can save drafts during the verification window — drafts are invisible to buyers and auto-submit once verification completes. Sellers see every listing's state at all times.

US-2.1 — Create a Listing Draft

As a: logged-in student I want to save my work-in-progress as a draft so that I don't lose it while composing or while I'm waiting for verification.

Acceptance Criteria:

- Given I am a logged-in student (verified or not) creating or editing a listing, when I choose to save as draft even with required fields missing, then the system saves the draft without showing validation errors for required fields.
- Given I am an unverified student, when I save a listing as a draft, then the draft is flagged with status 'pending verification'.
- Given I am an unverified student with one or more completed drafts, when my account becomes verified, then each flagged draft is automatically submitted for AI scoring without any manual action from me.
- Given I am the owner of a saved draft, when I open it from my drafts list, then I can edit any field and save it again as a draft.
- Given I am the owner of a saved draft, when I choose to delete it, then the draft is permanently removed and no longer appears in my drafts list.

US-2.2 — Submit Listing for Review

As a: verified student I want to submit a draft listing into the review pipeline so that buyers can discover and buy it.

Acceptance Criteria:

- Given I am a verified student, when I submit a draft and required fields are missing, then the system shows validation errors and does not enter AI scoring.
- Given I am a verified student, and I submit a complete draft, when submission completes, then the listing enters AI scoring and I am notified of the resulting state: live, live with reduced visibility, or under review.
- Given I open an existing draft and submit it, when submission is successful, then the original draft is consumed by the submission.

US-2.3 — AI Score Listing (System)

As the: system I want to score every submitted listing for risk so that low-risk listings go live, medium-risk listings go live with reduced visibility, and high-risk listings route to admin review.

Acceptance Criteria:

- Given a submitted listing receives a low-risk score, when scoring completes, then the listing is published with full visibility.
- Given a submitted listing receives a medium-risk score, when scoring completes, then the listing is published with reduced visibility (e.g. lower search rank, hidden from recommendations) and queued for non-urgent admin review.
- Given a submitted listing receives a high-risk score, when scoring completes, then the listing is not published, is held in the admin review queue, and that queue is prioritised over medium-risk background reviews.
- Given a listing has been scored, when scoring finishes, then the system logs: timestamp, risk score, flagged reasons, and the model version used.

US-2.4 — Manual Review

As an: admin I want to review AI-flagged listings so that legitimate ones reach buyers and problematic ones don't.

Acceptance Criteria:

- Given a listing is in the admin review queue, when I open it, then I see the listing details, photos, AI score, AI-flagged reasons, and the seller's history.
- Given I approve a flagged listing, when I confirm, then the listing is published and the seller is notified.
- Given I reject a flagged listing, when I confirm, then I must select a reason and add a free-text note; the listing is removed from the platform and the seller is notified with the reason.

- Given I request changes on a flagged listing, when I confirm, then I must provide a reason; the listing is returned to the seller as a draft (not live) accompanied by the reason, and the seller can edit and resubmit.
- Given I make any decision (approve, reject, request changes), when I confirm, then the system logs: admin ID, timestamp, outcome, and the reason text provided.

US-2.5 — Edit Active Listing

As a: seller I want to edit one of my live listings so that I can correct or update it.

Acceptance Criteria:

- Given my listing is reserved (a buyer has an active reservation), when I attempt to edit it, then I can only modify non-material fields (typo fixes, description clarifications); price, photos, and condition are locked.
- Given I edit a live listing (whether reserved or not), when I save the edit, then the listing is re-scored by AI and may result in reduced visibility or admin review; existing reservations remain valid and are not cancelled.
- Given I save any edit, when the edit is applied, then the system logs the change (timestamp, old value, new value, my user ID).

US-2.6 — Remove Listing

As a: seller I want to remove a listing from circulation so that it no longer appears on the marketplace.

Acceptance Criteria:

- Given I am removing a listing, when the removal flow begins, then I am presented with a reason picker (sold off-platform, no longer for sale, other).
- Given my listing is currently reserved by a buyer, when I attempt to remove it, then I must confirm twice (checkbox + confirm button), and the buyer is automatically notified of the removal.

US-2.8 — View All Listings

As a: seller I want to see all my listings in one place so that I can manage my inventory.

Acceptance Criteria:

- Given I am viewing my listings, when I apply a state filter (draft, live, reduced visibility, under review, removed), then only listings in the selected state are shown.
- Given I am viewing my listings, then each listing displays its state, AI score (if available), admin notes (if any), edit history, and reservation history.

- Given a listing is in a non-obvious state (e.g. reduced visibility, under review), when I view it, then I see an explanation of the state and any next steps.
- Given I am viewing my listings, when I apply a sort, then listings are ordered by the chosen criterion (date, state, price, expressions of interest).

Epic 3: Discovery and Recommendations

Help verified students find the materials they need — through casual browsing, intentful search, behaviour-driven recommendations, and a degree-aware planner that anticipates what they'll need across their studies. Surface seller reputation alongside listings so buyers can make informed decisions before expressing interest.

US-3.1 — Browse Listings

As a: student I want to browse the marketplace without a specific query so that I can casually explore what's available.

Acceptance Criteria:

- Given I access the marketplace without a search query, then I see a paginated or scrollable list of active listings.
- Given I am viewing browse results, then listings are sorted by relevance to my profile.
- Given I am viewing browse results, then each listing displays title, price, condition, primary photo, seller name, and seller rating summary.
- Given I am viewing browse results, when I apply filters (category, price range, condition, course/module, university) or sorting (newest, price asc/desc, relevance), then results update accordingly.
- Given I am an unverified student, when I browse listings, then I can view listings and I see a banner indicating limited access.

Epic 4: Reservation and Transaction

Enable verified students to reserve listings, coordinate and complete in-person meetup transactions, and confirm completion through a PIN-based closure system. Payment is processed at the point of physical inspection via PayFast. No refunds are issued — physical inspection before payment is the buyer's protection.

US-4.1 — Reserve a single item

As a: verified buyer I want to reserve a listing I'm interested in so that it is held for me while I arrange the meetup with the seller.

Acceptance Criteria:

- Given I am a verified student viewing an active, unreserved listing, when I express interest, then a reservation is created, the listing is marked as reserved, and the seller receives an automated initial message on my behalf.
- Given a reservation has been created, when it is active, then the listing is no longer available for other buyers to reserve.

- Given a reservation exists, when neither party has progressed, then the system automatically cancels the reservation, and the listing returns to active circulation.
- Given a reservation is auto-cancelled, when the cancellation fires, then both the buyer and seller are notified in-app.
- Given I am an unverified student, when I attempt to reserve a listing, then the action is blocked and I am shown a message explaining that verification is required.

US-4.2 – Reserve seller bundle from degree list

As a: *verified buyer, I want to reserve all relevant listings from a single seller that match my degree list so that I can arrange a bulk purchase in one meetup.*

Acceptance Criteria:

- *Given I confirm a bundle from a specific seller, when I proceed with reservation, then a reservation is created that includes all selected, available listings from that seller.*
- *Given some listings from that seller are already reserved, when I create the bundle reservation, then only unreserved listings are included, and I am informed which items were unavailable.*
- *Given a bundle reservation is created, then the seller is notified that this is a bundle purchase request and the reservation is treated as a single grouped transaction.*
- *Given a bundle reservation is active, then it follows a 7-day reservation window instead of the 24-hour reservation logic.*
- *Given a bundle reservation exists, then all items are grouped under one reservation context and cannot be partially confirmed through separate flows.*

US-4.3 – Cancel reservation (buyer-initiated)

As a: *buyer I want to cancel a reservation I created so that the seller's listing returns to circulation if I am no longer interested.*

Acceptance Criteria:

- *Given I have an active reservation, when I chose to cancel it, then the reservation is removed, the listing returns to active, and the seller is notified.*
- *Given I cancel a reservation, when the cancellation is confirmed, then the system records the cancellation timestamp and the initiating party for reputation tracking purposes.*

US-4.4 – Release reservation (seller-initiated)

As a: seller I want to release a reservation when the buyer is unresponsive so that my listing returns to active circulation.

Acceptance Criteria:

- Given a reservation exists on my listing, when I attempt to release it, then the release is only permitted if the buyer has been inactive for at least 12 hours since the reservation was created.
- Given the 12-hour cooldown has not elapsed, when I view the reservation, then the release option is disabled, and I see a message indicating how much time remains before I can release.
- Given I release a reservation, when confirmed, then the listing returns to active, the buyer is notified, and the event is logged.

US-4.5 – Purchase single item

As a: buyer with an active reservation I want to pay for an item at the meetup after inspecting it so that the transaction is completed only when I am satisfied.

Acceptance Criteria:

- Given I have met the seller and inspected the item, when I choose to proceed with payment, then I am directed to the PayFast payment flow with the correct amount pre-populated.
- Given payment via PayFast is successful, when the payment is confirmed, the system generates a secure 6-digit transaction PIN and delivers it to the seller via real-time WebSocket push.
- Given payment fails, when the PayFast flow returns a failure, then I am shown a clear error message and can retry without the reservation being cancelled.
- Given payment is successful, when the transaction is recorded, then the system stores: transaction ID, buyer ID, seller ID, item ID, amount, and timestamp — no card or bank details are stored.

US-4.6 – Confirm receipt (PIN verification)

As a: buyer, I want to enter the 6-digit PIN shown to me by the seller after physically inspecting the item, so that I can verify receipt and officially complete the purchase.

Acceptance Criteria:

- Given the buyer has completed payment, when the PayFast flow succeeds, then a PIN is generated and displayed to the buyer in-app.
- Given the buyer has shown me their PIN, when I enter it correctly in-app, then the transaction is marked complete and both parties are prompted to leave a review.

- Given I enter an incorrect PIN, when I submit, then I see an error and can retry; after a configured number of failed attempts the PIN is invalidated, and the case is flagged for support.
- Given the transaction is completed via PIN, when confirmed, then the systems log the completion timestamp, the confirming party (seller), and triggers the post-transaction review prompt for both parties.

US-4.7 – System auto-cancel reservation

As the: system I want automatically cancel the stale reservations so that listings do not remain locked when buyers or sellers are unresponsive.

Acceptance Criteria:

- Given a reservation has been active for 24 hours with no meaningful progression (no seller replies to the automated message, or no buyer response after seller reply), when the threshold is crossed, then the reservation is automatically cancelled, and the listing returns to active.
- Given an auto-cancellation fires, when it is executed, then both parties receive an in-app notification, and the event is logged with a reason of auto-expired.

US-4.8 – View reservation and transaction history

As a: student (buyer or seller) I want to see all my past and active reservations and completed transactions so that I can track my activity on the platform.

Acceptance Criteria:

- Given I am a verified student, when I navigate to my reservation/transaction history, then I see a list of all active reservations and completed transactions, with state, item, counterparty name, and date.
- Given I select a specific transaction, when I open it, then I see the full details: item, amount, meetup outcome, PIN confirmation status, and any associated review.
- Given I have an active reservation, when I view it, then I can see the time remaining before auto-cancellation and any relevant next steps.

Epic 5: Disputes and Moderation

Provide the channels through which users raise concerns about the marketplace and admins act on those concerns or on system-detected patterns. Disputes are limited to what the platform can reliably arbitrate — pre-payment listing quality complaints — and remedies are reputational, not financial. Automatic reputation flags can be challenged through a lightweight mechanism. Admin actions can be appealed through a structured process with a different reviewer, time limits, and bounded outcomes.

US-5.1 – File listing complaint

As a: buyer who attended a meetup, I want to file a complaint about a listing so that misleading or inaccurate listings can be acted on.

Acceptance Criteria:

- Given I attended a meetup for a listing, when I attempt to file a complaint, then it is accepted only within the configured time window after the meetup, and submissions outside this window are rejected.
- Given I am filing a complaint, then I must provide either: photos taken in-app at the meetup, or a refusal flag indicating the seller declined photos—and I cannot submit the complaint without one of these.
- Given I provide photos as evidence, then photos must be captured using the in-app camera, each photo includes a timestamp, and gallery uploads are not allowed.
- Optional: buyer may include a written description of the discrepancy.
- Given a complaint is successfully submitted, then it is routed to the admin dispute queue and includes reservation details, listing, evidence (photos/refusal flag), and chat history.
- Given a complaint has been reviewed, then the buyer is notified of the outcome.

US-5.2 – Dispute reputation flag

As a: user who has received a reputation flag, I want to dispute it so that incorrect or unfair flags can be reviewed.

Acceptance Criteria:

- Given I have received an automatic reputation flag, when I view my account status, then I can dispute the flag within the configured time window, and the option is not available after the window expires.
- Given I choose to dispute a flag, then I must provide a brief written explanation, and I can submit without additional evidence.
- Given a dispute is submitted, then it is queued for admin review, and the admin either removes the flag or lets it stand, and the decision is final.
- Given I submit disputes excessively within a defined period, then my access to the dispute mechanism is suspended.

US-5.3 – Appeal decision

As a: student who received an admin-issued action (strike, suspension, removal) I want to appeal the decision so that I have recourse when an admin has misread evidence or new information has come to light.

Acceptance Criteria:

- Given I received an admin-issued action, when I attempt to appeal, then I can submit an appeal within 7 days of the action, I cannot appeal the same action more than once, and appeals outside this window are not allowed.
- Given I am submitting an appeal, then I must provide either new evidence or a valid explanation of how the original decision misread existing evidence, and appeals based on disagreement alone are rejected.
- Given an appeal is submitted, then it is assigned to a different admin than the original decision-maker.
- Given an appeal is reviewed, then the outcome is one of: uphold, reduce penalty, or overturn, and the penalty cannot be increased.
- Given an appeal decision is made, then the user is notified with reasoning and the decision is final.

US-5.4 – Resolve dispute

As an: admin I want to review and resolve filed disputes so that legitimate complaints result in seller strikes and frivolous ones are dismissed.

Acceptance Criteria:

- Given I am viewing the dispute queue, then disputes are ordered by age (oldest first) and time-sensitive disputes are prioritised.
- Given I open a dispute, then I see the original listing, buyer evidence (photos or refusal flag), any description, chat history, and both users' reputation history.
- Given I review a dispute, then I can choose to uphold, dismiss, or request more information; and if upheld, the seller receives a strike and the listing is removed, and the buyer is notified of the outcome.
- Given I request more information, then the buyer is prompted to provide clarification, and the dispute remains open until resolved.
- Given a buyer repeatedly submits dismissed disputes within a defined period, then this contributes to a buyer-side reputation flag.
- Given a dispute is resolved, then the decision, reasoning, admin ID, and timestamp are recorded, and decisions follow the defined standard of proof.

US-5.5 – Review flagged users

As an: admin I want to see users surfaced by automatic pattern detection so that I can decide whether to act on cumulative behaviour that no individual incident warranted.

Acceptance Criteria:

- Given I am viewing the flagged-users dashboard, then I see users surfaced by automatic pattern detection when thresholds are met.
- Given I open a flagged user entry, then I see the user, triggered pattern type, underlying events with timestamps and source links, and full account history.

- Given I am viewing the dashboard, when I apply filters, then I can filter by pattern type and severity, and users with multiple overlapping patterns are surfaced higher.

US-5.6 - Act on user

As an: admin I want to apply a consequence to a user based on a dispute outcome or flag pattern so that the marketplace is protected from bad actors.

Acceptance Criteria:

- Given I am reviewing a user case (dispute outcome or flag pattern), when I choose to act, then I can apply one of: warning, strike, temporary suspension, or permanent removal.
- Given I am applying an action, then I must select a reason and provide a free-text note.
- Given an action is applied to a user, then the user is notified via in-app and email, and the notification includes the action type, reason, and duration where applicable.
- Given a user receives an action, then it is appealable according to appeal rules, and permanent removals due to fraud or POPIA-mandated takedowns are not appealable.
- Given an action is taken, then it is logged with admin ID, timestamp, user, action type, reason, and source case.
- Given a user accumulates strikes, then threshold rules trigger escalation conditions (e.g. review for suspension).

US-5.7 – Review appeal

As an: admin (different from the one who made the original decision) I want to review an appealed decision so that users have meaningful recourse against admin error.

Acceptance Criteria:

- Given I am viewing the appeal queue, then appeals are ordered by age (oldest first).
- Given I open an appeal, then I can see the original action, original admin reasoning, user appeal grounds, and all case evidence.
- Given I am assigned to review an appeal, then I cannot be the same admin who issued the original decision.
- Given I complete an appeal review, then I must provide written reasoning visible to the user, and the user is notified of the outcome.

Epic 6: Meetup Logistics

Provide students with map-based support for meetups arranged in chat, helping them find the agreed location and navigate there. Map functionality is provided via Leaflet and OpenStreetMap, utilizing the Nominatim engine for campus landmark geocoding.

Out of scope: chat, notifications, presence verification, location sharing during transit (privacy-preserving check-in is handled in package 4.)

US-6.1 – View meetup location on map

As a: student with an arranged meetup I want to see the agreed meetup location on a map so that I know where to go.

Acceptance Criteria:

- Given a meetup location has been agreed and captured in chat, then it is available in the related reservation/transaction view.
- Given I open a reservation with a meetup location, then I can view the location on an interactive in-app map and the meetup point is clearly marked.
- Given I am viewing the meetup map, then I can see surrounding landmarks, and I can see my current location for orientation.

US-6.2 – Pre-meetup attendance check-in

As a: student with an upcoming meetup, I want to confirm my attendance before the meetup so that the system can verify participation for audit purposes.

Acceptance Criteria:

- Given a meetup is scheduled, when it is 15 minutes before the meetup time, then both participants receive a check-in prompt.
- Given I receive a check-in prompt, then I can either confirm attendance or decline participation.
- Given I confirm attendance, then my confirmation is recorded for audit purposes.
- Given I decline attendance, then my participation is marked as forfeited according to the system rules.
- Given no response is given before the meetup time, then the system records the attendance as unconfirmed.

3. Use Case Diagrams

UC1: User and Identity Management

UC1.1: Register Account

- **TUCBW** a prospective student, not registered in the system yet and has a valid South African university email address, submits the registration form with all the required information.
- **TUCEW** the user account is created, but the user is placed in a “Pending verification” state (unverified). An OTP is sent to the provided student email.

UC1.2: Get OTP

- **TUCBW** the user has successfully registered and is on the OTP verification page.
- **TUCEW** a One-Time Password (OTP) is generated and sent to the user’s registered email address.

UC1.3: Verify OTP

- **TUCBW** the user has received the OTP via email and has pressed verify on the verification screen.
- **TUCEW** the entered OTP is validated. The user moves to the next step of the verification workflow (Uploading Proof of Registration).

UC1.4: Upload Proof of Registration

- **TUCBW** the user successfully verified their email via OTP, and the user is on the upload Proof of Registration screen.
- **TUCEW** the Proof of Registration file is uploaded to the system, and the verification process is initiated.

UC1.5: AI Verify Proof

- **TUCBW** the user has successfully uploaded their Proof of Registration document.
- **TUCEW** the AI processes the Proof of Registration and outputs a confidence score, which determines the next action.

UC1.6: Manual Verification

- **TUCBW** a user’s account or listing has been added to the admin queue (due to Medium/Low AI confidence, Listing Risk, or Manual Report).

- **TUCEW** The admin makes a decision (Approve/Reject/Request Changes), updating the user's status or listing visibility.

UC1.7: Resubmit Proof of Registration

- **TUCBW** the user has been flagged for manual review or rejected, and the admin has requested a new/updated proof of registration.
- **TUCEW** the new file is uploaded to the system; the user status is reset to "Pending verification".

UC1.8: Login

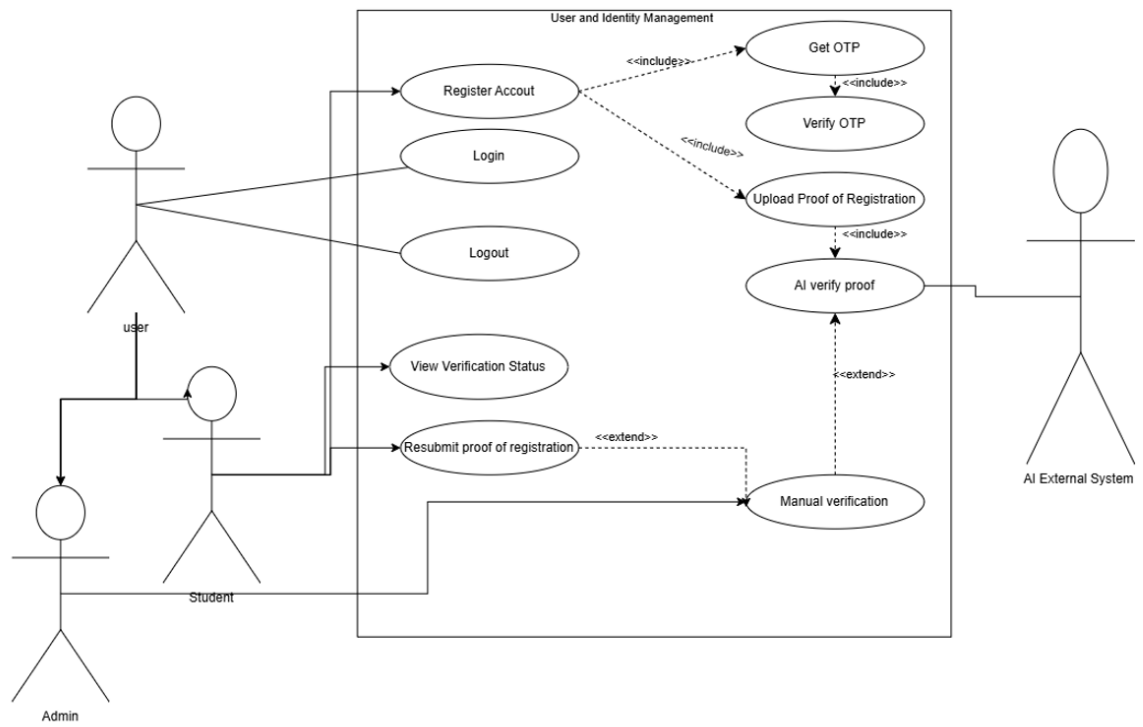
- **TUCBW** a registered student enters their valid credentials (email and password) and press Login.
- **TUCEW** the user is granted access to the system. Access permissions are determined by their verification status (e.g., Partial Access if unverified, Full Access if verified).

UC1.9: View Verification Status

- **TUCBW** a logged in user navigates to their profile page.
- **TUCEW** the user sees their current verification status ("Pending", "Verified", "Rejected" or "Review Required").

UC1.10: Logout

- **TUCBW** the user who is currently logged into an active session, presses Logout from their profile page or the sidebar.
- **TUCEW** the user's session is terminated, and the user is redirected to the Login screen.



UC2: Listing and Inventory Management

UC2.1: Upload a Listing

- **TUCBW** the seller is logged in, and they are on the upload listing page, then they fill in the listing details (title, description, price, condition, course tags), upload photos and press Submit.
- **TUCEW** the listing is saved as draft (invisible to buyers), if the seller is verified, the system automatically proceeds to the AI risk-scoring pipeline. If the seller is unverified, the listing remains as a draft until they are verified.

UC2.2: Edit Active Listing

- **TUCBW** the seller has an existing listing that is currently live (visible to buyers), and the seller presses Edit Listing in the Listings' table.
- **TUCEW** the changes are saved and reflecting in the listings' table.

UC2.3: Save Listing Draft

- **TUCBW** the seller has entered only the information available to them and pressed save as draft.
- **TUCEW** the partially filled listing data persisted as a draft in the system, and the listing is reflecting on the listings' table with a draft status.

UC2.4: Remove Listing

- **TUCBW** the seller has pressed the delete icon on an of the listing in their listings' table.
- **TUCEW** the listing is permanently deleted from the platform, and it is removed from the buyer's browse listing page (if it was live).

UC2.5: View all Listings

- **TUCBW** the seller is logged into the system and navigates to the My Listings Page.
- **TUCEW** the system displays all the seller's uploaded listings.

UC2.6: Acknowledge Reservation

- **TUCBW** the buyer has reserved the item, and the seller sees the reservation on their Reserved page and pressed acknowledge.
- **TUCEW** the chat thread has opened, and the buyer and seller can chat back and forth.

UC2.7: Cancel Reservation

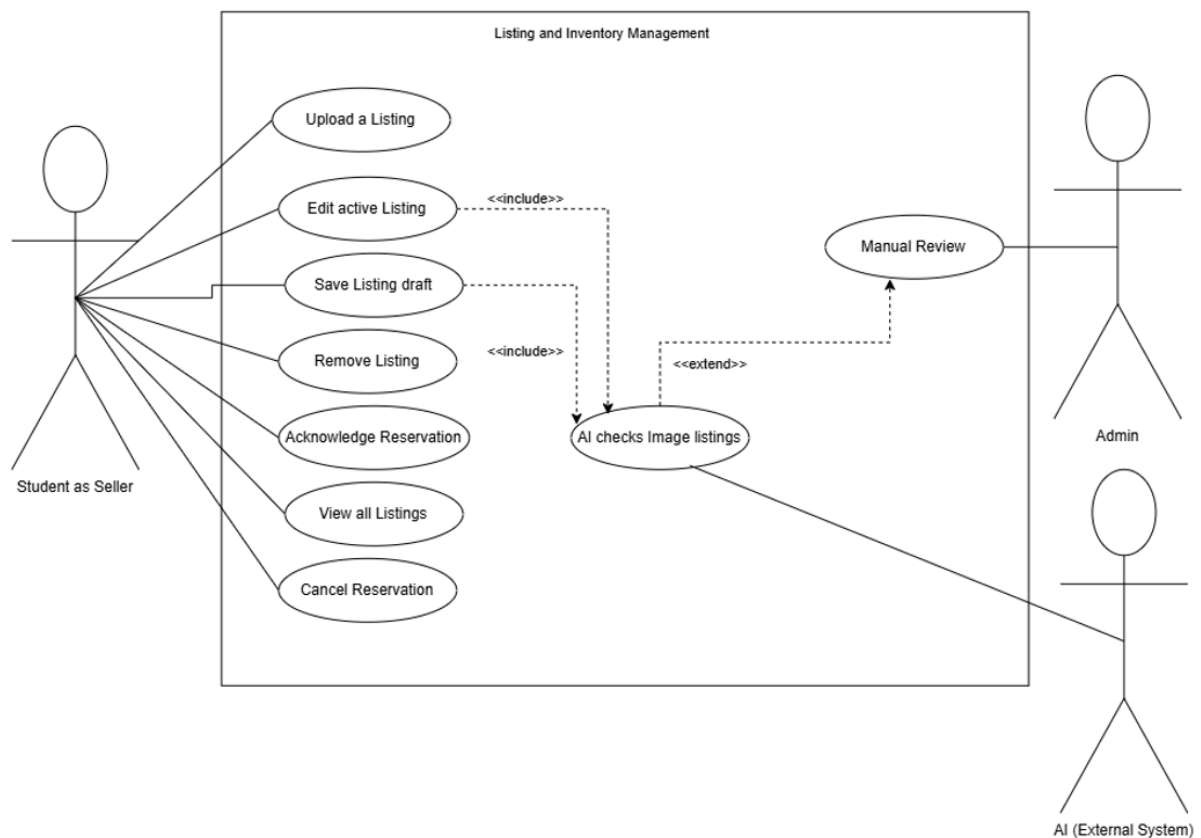
- **TUCBW** the seller cancels an active reservation after 12 hours of buyer silence.
- **TUCEW** the reservation is voided. The listing is immediately released back to the marketplace, making it available for other buyers to reserve.

UC2.8: AI checks Image listing

- **TUCBW** the seller has uploaded a listing.
- **TUCEW** the AI processes the listing photos and outputs a confidence score, which determines the next action.

UC2.9: Manual Review

- **TUCBW** the seller's listing is in the Admin Listing queue (due to Medium/Low AI confidence)
- **TUCEW** the admin makes a decision (Approve/Reject/Request Changes), updating the listing's status and visibility.



UC3: Discovery and Recommendations

UC3.1: View dashboard

- **TUCBW** the student (Guest or Verified) is logged in and navigates to the main marketplace feed.
- **TUCEW** the dashboard loads, displaying all published/live listings.

UC3.2: Generate suggested materials

- **TUCBW** the student is logged in and on the Dashboard.
- **TUCEW** the system displays a personalized list of recommended listings (e.g., “Popular for COS151 at University X” or “Frequently bought together”).

UC3.3: Search Listing

- **TUCBW** the student enters a keyword (title) into the search bar on the marketplace.
- **TUCEW** the system returns a list of live listings matching the query.

UC3.4: Sort Results

- **TUCBW** the student has performed a Search or is viewing a category and selects a sorting option.
- **TUCEW** the displayed list of listings is re-ordered according to the selected criteria.

UC3.5: Filter Results

- **TUCBW** the student is viewing the marketplace or search results and applies filter criteria.
- **TUCEW** the system narrows down the listing results to show only the items matching the selected filters.

UC3.6: View Material Details

- **TUCBW** the student clicks on a specific listing card from the Browse All Listings page, search results, or Wishlist.
- **TUCEW** the system shows the listing’s details.

UC3.7: Add to Wishlist

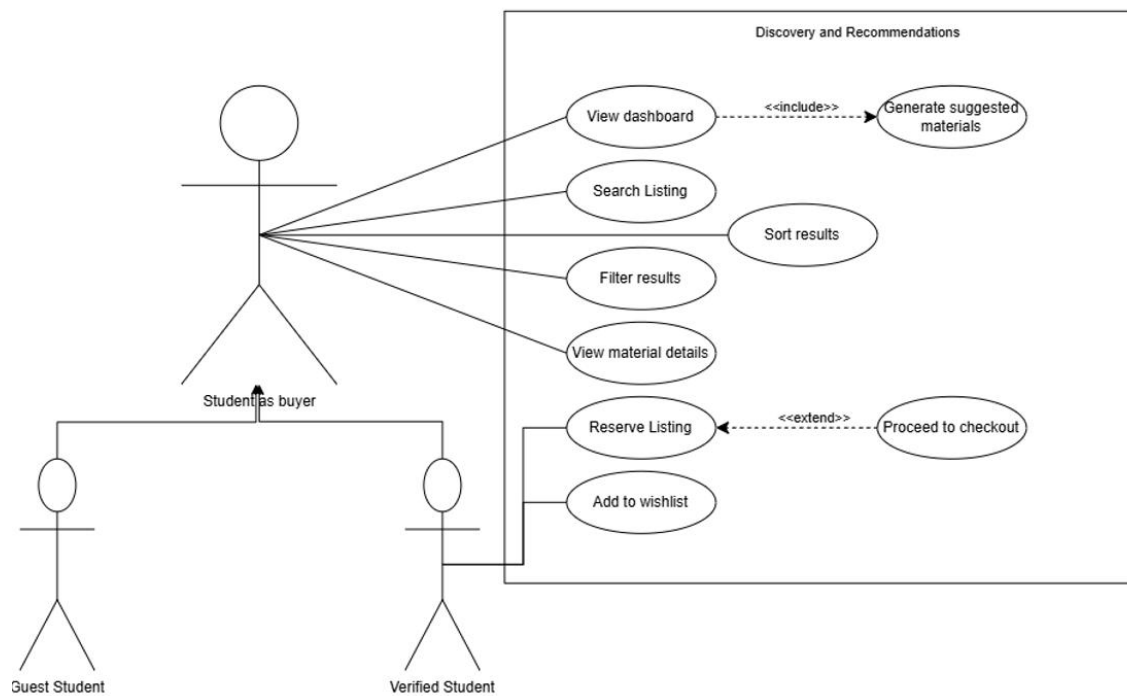
- **TUCBW** a logged in user is viewing the details of a live listing, and presses “Add to Wishlist”.
- **TUCEW** the listing is saved to the student’s personal Wishlist for future quick access.

UC3.8: Reserve Listing

- **TUCBW** the verified student presses “Reserve” on the available listing’s card.
- **TUCEW** the system creates a temporary reservation, the 24-hour timer starts, and an automated initial “interested” message is sent to the seller via the chat.

UC3.9: Proceed to Checkout

- **TUCBW** a verified student with an active reservation, has coordinated a meetup with the seller clicks “Pay” on the Meetup Details Page.
- **TUCEW** the buyer is redirected to the Payfast payment gateway to complete their payment.



UC4: Reservation and Transaction

UC4.1: Reserve Item

- **TUCBW** the verified student/buyer views a listing and selects “Reserve”
- **TUCEW** an automated initial “interested” message is generated and sent to the seller, opening a chat thread and placing a 24-hour initial reservation timer on the item.

UC4.2: Cancel Reservation

- **TUCBW** the buyer manually chooses to cancel their active reservation, presses “Cancel” on the reservation card.
- **TUCEW** the reservation is removed, and the item becomes available to other buyers again.

UC4.3: Reserve full degree list

- **TUCBW** the buyer chooses to reserve a bundle/full degree list of textbooks from a single seller.
- **TUCEW** a 7-day reservation window is initiated for the bundle, and the seller is notified.

U4.4: Purchase item

- **TUCBW** the buyer attends the meetup, inspects the physical item, approves its condition, and initiates payment.
- **TUCEW** the payment is redirected to the payment gateway (PayFast) to process.

UC4.5: Process payment

- **TUCBW** the payment flow is initiated via PayFast.
- **TUCEW** the PayFast backend sends a successful webhook response to the system, verifying the payment and generating a confirmation PIN for the buyer.

UC4.6: Confirm Receipt

- **TUCBW** the seller shows the generated PIN to the buyer at the meetup.
- **TUCEW** the buyer enters the PIN into the app, closing and completing the transaction.

UC4.7: Raise Dispute

- **TUCBW** a buyer identifies a problem and submits a report/flag.

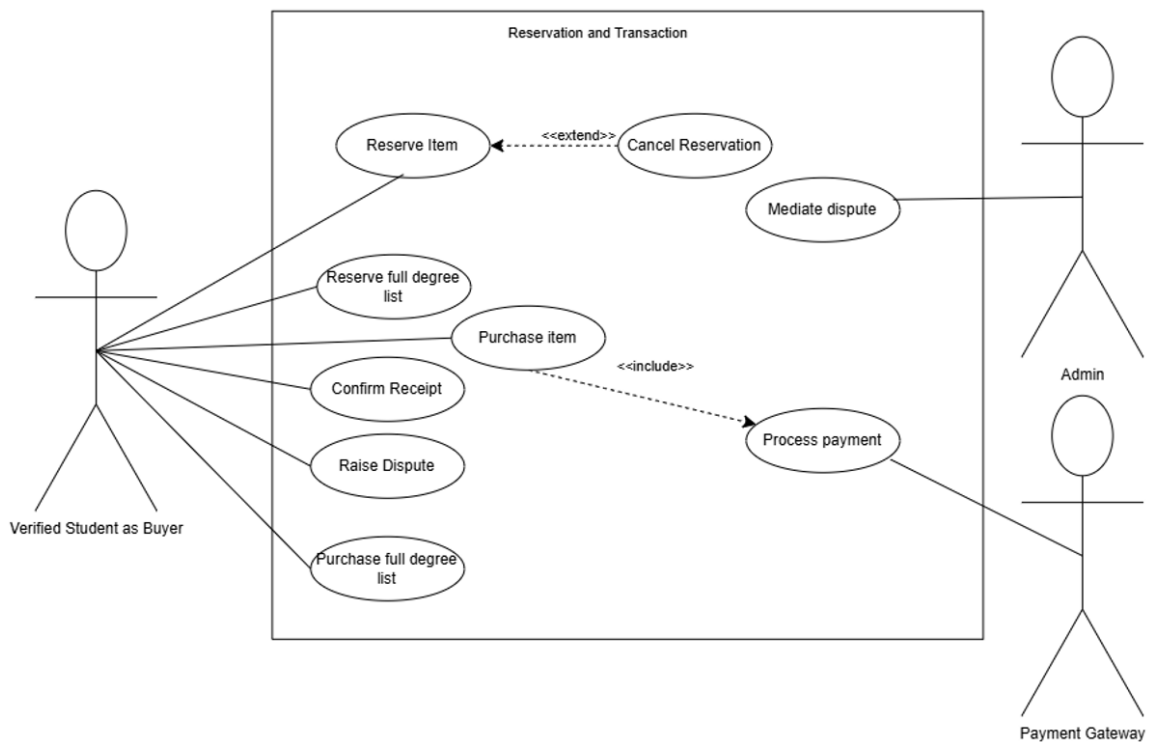
- **TUCEW** the complaint/evidence is compiled into a case record for processing by an admin.

UC4.8: Purchase full degree list

- **TUCBW** a buyer reserves a full degree list from a seller.
- **TUCEW** a 7-day timer is started for the reservation, and a chat thread is also open where the buyer can communicate with the seller.

UC4.9: Mediate dispute

- **TUCBW** an admin opens a reported/flagged dispute case.
- **TUCEW** and admin reaches a decision, applying a seller strike, listing removal, or account action.



UC5: Admin and Dispute Resolution

UC5.1: View pending verifications

- **TUCBW** the admin accesses the verification queue containing students' Proof of Registration.
- **TUCEW** an admin selects a specific student verification case to inspect.

UC5.2: Approve/reject proof of registration

- **TUCBW** the admin evaluates an escalated Proof of Registration document.
- **TUCEW** the admin makes a decision, and the user is notified via email.

UC5.3: View AI failed listings

- **TUCBW** the admin opens the moderation queue for high-risk or flagged marketplace listings.
- **TUCEW** the admin loads the details, images, seller history, and AI risk reasons for a flagged listing.

UC5.4: Manually approve/reject materials

- **TUCBW** the admin reviews a flagged/high-risk listing.
- **TUCEW** the admin selects Approve (listing goes live), Reject (listing removed, seller notified), or Request changes (listing sent back to draft).

UC5.5: View dispute cases

- **TUCBW** the admin opens the dispute management dashboard.
- **TUCEW** the system displays all the compiled cases.

UC5.6: Review evidence

- **TUCBW** the admin opens a specific dispute file to examine available evidence.
- **TUCEW** the admin finishes evaluating all provided evidence and proceeds to decision-making.

UC5.7: Mediate and decide outcome

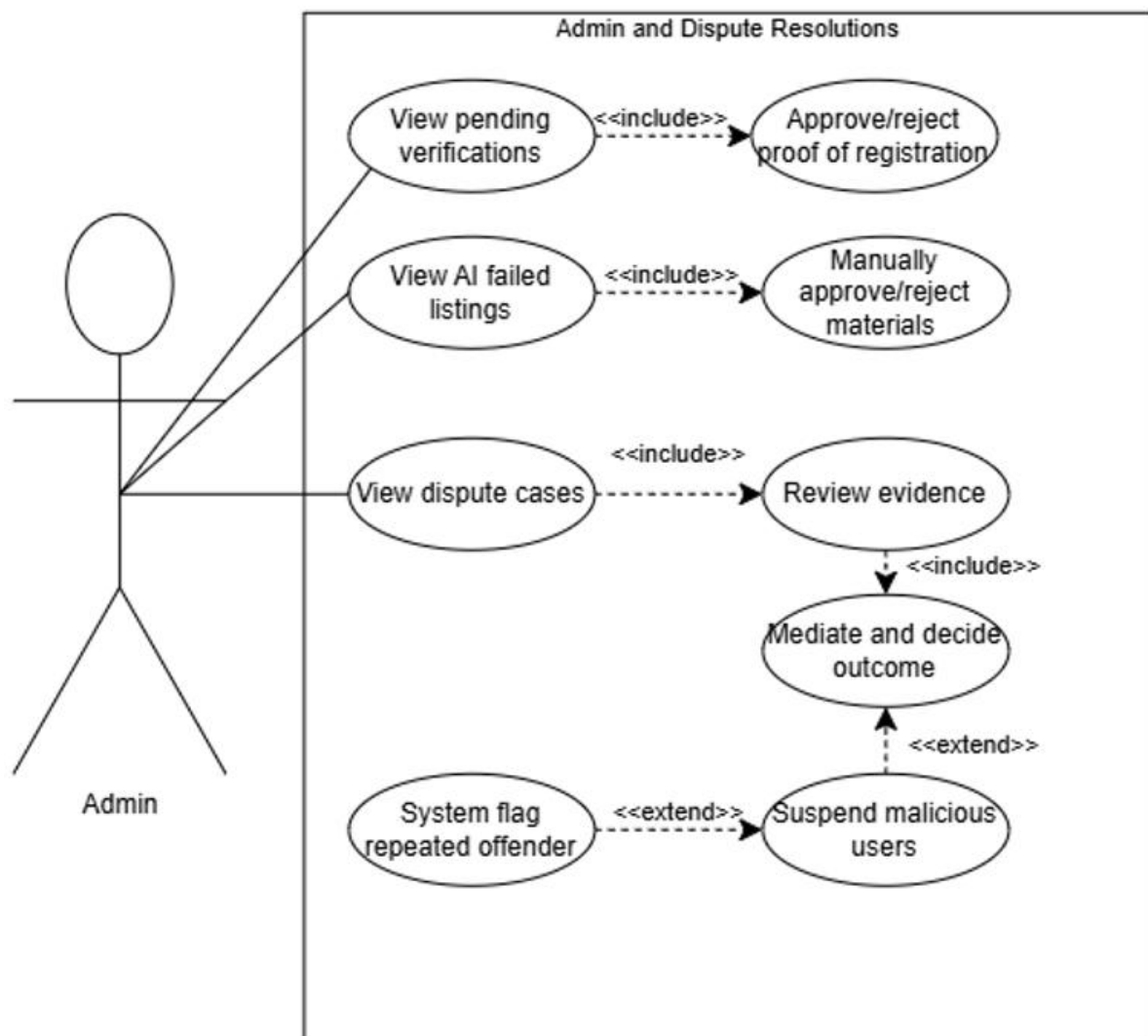
- **TUCBW** the admin weighs the evidence and user historical patterns against platform policies.
- **TUCEW** the admin issues a final verdict (e.g. strike assigned, listing removed, or dispute dismissed).

UC5.8: System flag repeated offender

- **TUCBW** the system records cumulative negative signals exceeding thresholds limits.
- **TUCEW** the system automatically logs severe risk flag against the account or triggers escalation.

UC5.9: Suspend malicious users

- **TUCBW** the repeated offender threshold is crossed, or an Admin manually triggers suspension after dispute mediation.
- **TUCEW** the malicious account is restricted/suspended from the platform, and trust (reputation score) are updated.



UC6: Communication and Logistics

UC6.1: Open chat thread

- **TUCBW** a buyer presses “Reserve” on the listing’s card of the listing’s details page; a seller acknowledges and click messages on their side bar.
- **TUCEW** a direct, private messaging channel between the buyer and seller is established.

UC6.2: Send message

- **TUCBW** a user types and sends text/coordination details in the chat thread.
- **TUCEW** the message is delivered to the recipient and rendered in the chat view.

UC6.3: Get push notification

- **TUCBW** an event occurs, 15-minute meetup check-in reminder, listing status change, or new chat message.
- **TUCEW** the system delivers a push notification to the user’s device.

UC6.4: Enter seller’s generated pin

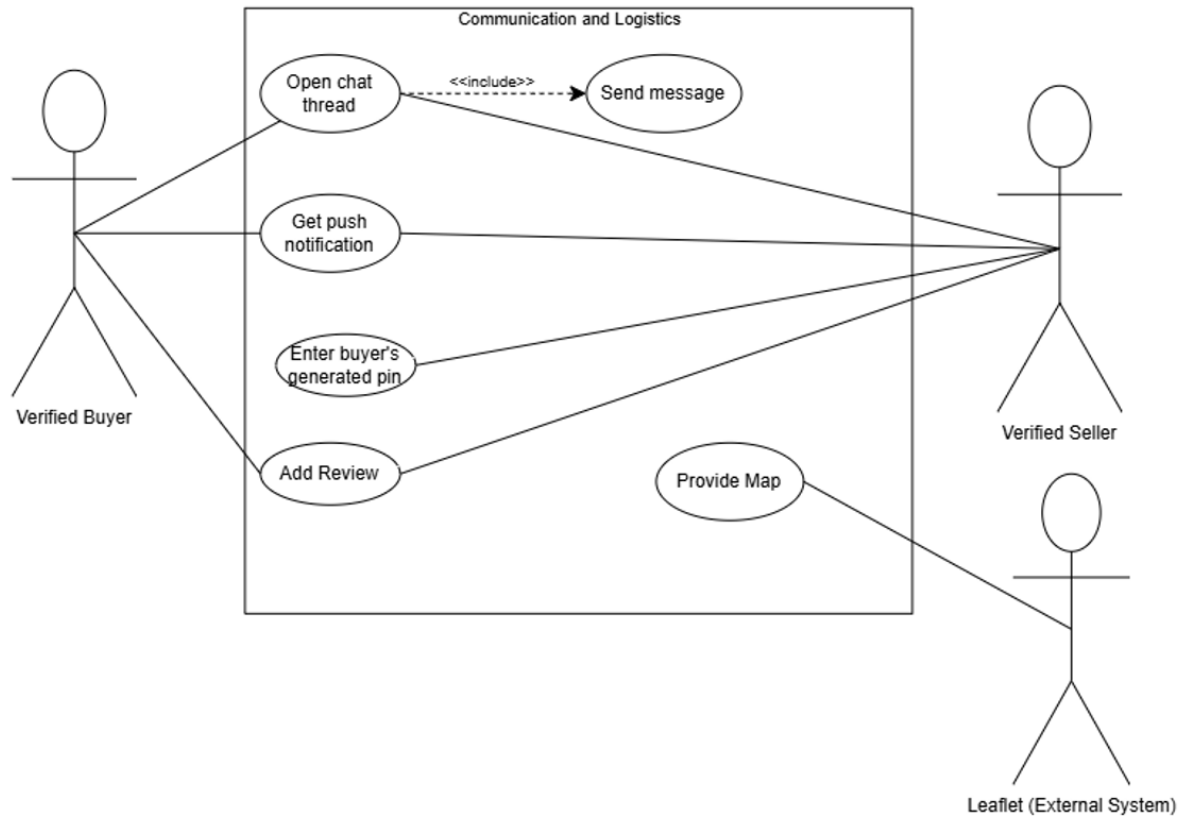
- **TUCBW** the buyer completes payment via PayFast, seller receives a system-generated PIN.
- **TUCEW** the buyer enters and submits the PIN, confirming successful handover and completing the transaction.

UC6.5: Add Review

- **TUCBW** a transaction is closed via PIN verification, and it appears on the buyer’s My Orders page and the seller’s My Sales page.
- **TUCEW** the buyer/seller submits a start rating and feedback for the transaction.

UC6.6: Provide Location

- **TUCBW** the buyer/seller taps “I’m here” within the 15-minute pre-meetup to 30-minute post-meetup window.
- **TUCEW** the app requests GPS coordinates via Leaflet, verifies proximity, and logs the verified check-in timestamp.



4.Functional Requirements

R1: User and Identity Management

R1.1: User Registration

R1.1.1 The system shall allow a registered university student to be registered in the system.

R1.1.2 The system shall provide an input area for student email, student number and proof of registration pdf.

R1.1.3 The system shall verify that the user is a student at said university by sending an OTP to the user's university email.

R1.1.4 The system shall allow users to resend OTP to their email if it expires.

R1.1.5 The system shall extract text from the uploaded proof of registration pdf and compare the student number and degree against the registration form data using Ai.

R1.1.6 The system shall assign a confidence score (0-100) to the ai verification result.

R1.1.7 The system shall provide manual verification of students' degrees if AI fails or gives a score below 75.

R1.1.8 The system must be able to provide users with a verification status.

R1.1.9 The system shall allow users to resubmit proof of registration pdf if manual verification fails.

R1.2: The system shall allow users to remain logged into the website or mobile application.

R1.3: The system shall allow Logged in users to reset their password.

R1.4: The system shall allow users to switch roles between being a seller or buyer.

R1.5: The system shall allow logged in users to log out.

R1.6: The system shall restrict email-verified users (OTP confirmed, proof or registration decision pending) to read-only access.

R1.7: The system shall prevent email-verified users (OTP confirmed, proof or registration decision pending) from making purchases.

R1.8: The system shall store listings from email-verified users (OTP confirmed, proof of registration decision pending) sellers as drafts.

R1.9: The system shall have a POPIA Purge Worker:

R1.9.1: The system shall run an automated background retention sweep every 24 hours.

R1.9.2: The system shall permanently delete student proof-of-registration binary payloads from the database when the administrative decision age exceeds 30 days.

R1.9.3: The system shall append an immutable audit record for every purged document.

R2: Listing and Inventory Management

R2.1: Create a Listing Draft

R2.1.1: The system shall allow users to upload images of materials.

R2.1.2: The system shall allow users to provide details of the product they are selling.

R2.1.3: The system shall use AI to assess uploaded images for object recognition (must identify as a book or educational material) before it is published.

R2.1.4: The system shall assign a quality score (0–100) to each image.

R2.1.5: The system shall automatically approve users when the AI confidence score exceeds the threshold.

R2.1.6: The system shall have an admin intervene to manually check quality if score is below threshold.

R2.1.7: The system shall apply different visibility rules based on risk level.

R2.2: The system shall allow sellers to edit their listings.

R2.3: The system shall allow sellers to delete their listings.

R2.4: The system shall allow sellers to view all their listings.

R2.5: The system shall allow sellers to save their listing so it's in the process of getting published.

R2.6: The system shall keep all their new listings in a draft state until submission.

- R2.7: The system shall prevent unverified sellers from submitting listings.
- R2.8: The system shall publish listings automatically for low-risk scores.
- R2.9: The system shall reduce visibility of medium-risk listings.
- R2.10: The system shall prevent high-risk listings from being published.
- R2.11: The system shall allow sellers to unreserve the listing the buyer has reserved on their material.

R3: Discovery and Recommendations

R3.1: Degree-Specific Dashboard

- R3.1.1:** The system shall provide users with suggested materials.

R3.2: Search

- R3.2.1:** Buyers can search by seller name.
- R3.2.2:** Buyers can search listings by material title.
- R3.2.3:** Buyers can search listings by course code.

R3.3: Filters

- R3.3.1:** Buyers shall be able to filter by category.
- R3.3.2:** Buyers shall be able to filter by condition.

R3.4: Sorters

- R3.4.1:** Buyers shall be able to sort results by date listed.
- R3.4.2:** Buyers shall be able to sort by price range.

R3.5: The system shall allow users to view details of a specific listing.

R3.6: The system shall allow buyers to reserve listing.

R3.7: The system shall allow buyers to add material to wish list.

R3.8: The system shall allow users to ask questions on a specific listing:

- R3.8.1:** The system shall allow verified buyers to post textual questions on any active listing.
- R3.8.2:** The system shall mask the inquirer's identity (e.g., initials only) to prevent off-platform transaction circumvention.

R3.8.3: The system shall allow the listing owner to submit a single public answer to each posted question.

R4: Reservation and Transaction

R4.1: Reservation Rules

R4.1.1: The system shall allow buyers to reserve items for a 24-hour window period.

R4.1.2: The system shall require seller acknowledgement within 24 hours.

R4.1.3: The system shall allow buyers to reserve the full degree list of materials.

R4.1.4: The system shall buyers to reserve the full degree package for days.

R4.1.5: The system shall allow sellers to manually release reservations after 12 hours of buyer inactivity.

R4.1.6: The system shall stop reservation timers when the meetup is scheduled.

R4.1.7: The system shall expire reservations if either party is inactive that is the meetup was never scheduled.

R4.2: The system shall provide a payment gateway to process buyers' payments.

R4.3: The system shall allow buyers to confirm payment once they've met up with the seller.

R4.3.1: The system shall generate a pin for the buyer.

R4.3.2: The system shall allow sellers to enter a pin given by the buyer during

their meetup.

R4.4: The system shall allow buyer to raise a dispute if materials were not as they were displayed.

R4.4.1: The system shall allow buyers to submit up to 50 listing IDs alongside an aggregate budget limit.

R4.4.2: The system shall execute a 0/1 Multiple-Choice Knapsack algorithm to determine the item selection that maximizes total items within the budget.

R4.4.3: The system shall automatically apply seller-configured bundle discounts.

R4.4.4: The system shall execute atomic grouped reservations for each participating seller.

R5: Admin and Dispute Resolution

R5.1: Verification Management

R5.1.1: The system shall allow admins to approve or reject proof of registration.

R5.2: Listing Review

R5.2.1: The system shall allow admins to manually reject or approve materials.

R5.3: Dispute Handling

R5.3.1: The system shall allow admins to review submitted evidence.

R5.3.2: The system shall allow admins to suspend any malicious users.

R5.3.3: The system shall allow admins to flag users as a warning.

R5.4: The system shall provide an admin interface to submit and store POPIA compliance documentation.

R6: Communication and Logistics

R6.1: Real-Time Chat

R6.1.1: The system shall allow users to open a chat thread.

R6.1.1.1: The system shall allow users to send messages.

R6.1.1.2: The system shall allow users to receive notifications.

R6.2: GPS Meetup

6.2.1: The system shall allow users to be able to select a meeting point from the provided map interface.

R6.2.2: The system shall allow either party to propose a meeting point; the other party must accept before it is confirmed.

R6.3: The system shall allow users to add review once a transaction has been completed.

R6.4: Mutual Availability

R6.4.1: The system shall parse uploaded iCalendar files to extract recurring weekly academic commitments.

R6.4.2: The system shall calculate the common free-time intersection between buyer and seller using 1D interval algebra bounded between 08:00 and 20:00.

R6.4.3: The system shall project conflict-free meetup windows over a 7-day rolling calendar horizon.

5. Non-Functional Requirements

QR1: Performance

QR1.1 (Read Latency): The system shall deliver read endpoints (feed, search, and browse) with a p95 response time of ≤ 800 ms and an error rate of $<1\%$ under baseline concurrent user loads.

QR1.2 (Real-Time Communication): The system shall deliver real-time chat messages to active recipients with a p95 latency of ≤ 500 ms using push-based protocols.

QR2: Scalability

QR2.1 (Load Scalability): Under 2x baseline load, the system shall maintain an rate of $<1\%$ and a p95 response latency non-exceeding 1.5x baseline p95, utilizing automatic horizontal scaling without requiring application code modifications.

QR3: Security

QR3.1 (Role-Based Access Control): The system shall enforce role-based access control, ensuring 100% of administration endpoints reject non-admin authentication tokens with an HTTP 403 Forbidden status.

QR3.2 (Payment Callback Integrity): The system shall verify the digital signature of incoming payment notifications prior to execution, immediately rejecting unsigned or tampered callbacks without modifying system state.

QR3.3 (Secure OTP Handling): The system shall store One-Time Passwords (OTPs) using a peppered cryptographic hashing scheme and validate them using constant-time comparison algorithms to prevent timing attacks.

QR3.4 (Vulnerability Exposure): The system shall contain zero High or Critical dynamic security vulnerabilities (DAST) on production/staging environments, with all medium findings documented and triaged.

QR3.5 (Dependency Vulnerability Scanning): The system build pipeline shall block deployment if any direct or transitive software dependency contains a High or Critical Common Vulnerabilities and Exposures (CVE) entry.

QR4: Reliability & Availability

QR4.1 (Webhook Idempotency): The system shall handle duplicate payment notifications idempotently using payment provider reference keys, producing exactly one system state change regardless of delivery attempts.

QR4.2 (Fault Isolation): The system's background processing architecture shall isolate individual worker cycle failures, preventing a single task error from terminating the background service.

QR4.3 (Deployment Rollback): The system release pipeline shall automatically roll back to the previous stable revision if a newly deployed instance fails automated health checks during deployment.

QR4.4 (System Availability): The system shall maintain an uptime availability of $\geq 99.5\%$ over the 30-day period preceding release milestones.

QR5: Maintainability & Quality

QR5.1 (Code Quality and Test Coverage): The codebase shall maintain $\geq 80\%$ automated test coverage and pass static analysis quality gates with zero blocker or critical issues before code merge.

QR5.2 (Database Migration Integrity): Automated schema migrations shall execute successfully and cleanly against a fresh, unpopulated database during CI pipeline verification.

QR6: Usability & Accessibility

QR6.1 (Accessibility Standard): Public-facing and administrative web interfaces shall present zero Serious or Critical accessibility violations under automated accessibility audits.

QR6.2 (Task Efficiency): The platform shall enable a user to complete defined core tasks end-to-end within ≤ 2 minutes.

QR7: Administration & Moderation

QR7.1 (Moderation Queue Ordering): The administrative moderation interface shall process and present pending verification cases strictly in First-In-First-Out (FIFO) chronological order based on submission timestamp.

QR7.2 (Audit Logging): Every administrative state-changing action shall generate an immutable audit log entry recording the actor ID, timestamp, and action reason.

QR7.3 (Decision Notification): The platform shall automatically dispatch an in-app/email notification to the target user immediately upon an administrative decision being made.

QR7.4 (SLA Breach Escalation): The administrative dashboard shall automatically flag and highlight any unresolved moderation case that exceeds the defined Service Level Agreement window (48-72 hours).

6. Domain Model

